

CS & IT ENGINEERING



Computer Network - 2

MAC Sub-Layer

Lecture No. - 04

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Recap of Previous Lecture



Topic

Slotted ALOHA

Topic

CSMA





Topics to be Covered



Topic

CSMA

Topic

CSMA/CD $\Rightarrow$ V.V. Imp.

Topic

Ethernet $\leftarrow$

[MCQ]

[GATE-IT-2004] [2 Mark]



#Q. Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability $p = 0.2$ in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. What is the maximum number of hosts which this protocol can support, if each host has to be provided a minimum through put of 0.16 frames per time slot?

- ☐ A 1
- ☒ B 2
- ☐ C 3
- ☐ D 4

Ans: B

Solution :-

Each host transmits in slot with probability (p) = 0.2

Maximum number of hosts (N) = ?

Throughput of Host = 0.16

$$\text{Throughput of Host} = p * (1 - p)^{(N-1)}$$

$$0.16 = 0.2 * (0.8)^{(N-1)}$$

[NAT]

[GATE-2015][2 Mark]



#Q. Consider a LAN with four nodes S1, S2, S3 and S4. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmit in the same slot. The probabilities of generation of a frame in a time slot by S1, S2, S3 and S4 are [0.1, 0.2, 0.3 and 0.4] respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _____.



Solution :-

Node S1 transmits in a slot with probability (P_1) = 0.1

Node S2 transmits in a slot with probability (P_2) = 0.2

Node S3 transmits in a slot with probability (P_3) = 0.3

Node S4 transmits in a slot with probability (P_4) = 0.4

Throughput of Node S1 = $(P_1) * (1 - P_2) * (1 - P_3) * (1 - P_4)$

Throughput of Node S2 = $(1 - P_1) * (P_2) * (1 - P_3) * (1 - P_4)$

Throughput of Node S3 = $(1 - P_1) * (1 - P_2) * (P_3) * (1 - P_4)$

Throughput of Node S4 = $(1 - P_1) * (1 - P_2) * (1 - P_3) * (P_4)$

$$\begin{aligned} \text{Throughput of Channel} &= \text{Throughput of Node S1} \\ &+ \text{Throughput of Node S2} \\ &+ \text{Throughput of Node S3} \\ &+ \text{Throughput of Node S4} \end{aligned}$$

$$\begin{aligned} \text{Throughput of Channel} &= (0.1 \times 0.8 \times 0.7 \times 0.6) \\ &+ (0.9 \times 0.2 \times 0.7 \times 0.6) \\ &+ (0.9 \times 0.8 \times 0.3 \times 0.6) \\ &+ (0.9 \times 0.8 \times 0.7 \times 0.4) \\ &= \boxed{0.4404} \end{aligned}$$



Topic : CSMA



→ Carrier Sense Multiple Access

→ Sense before transmit

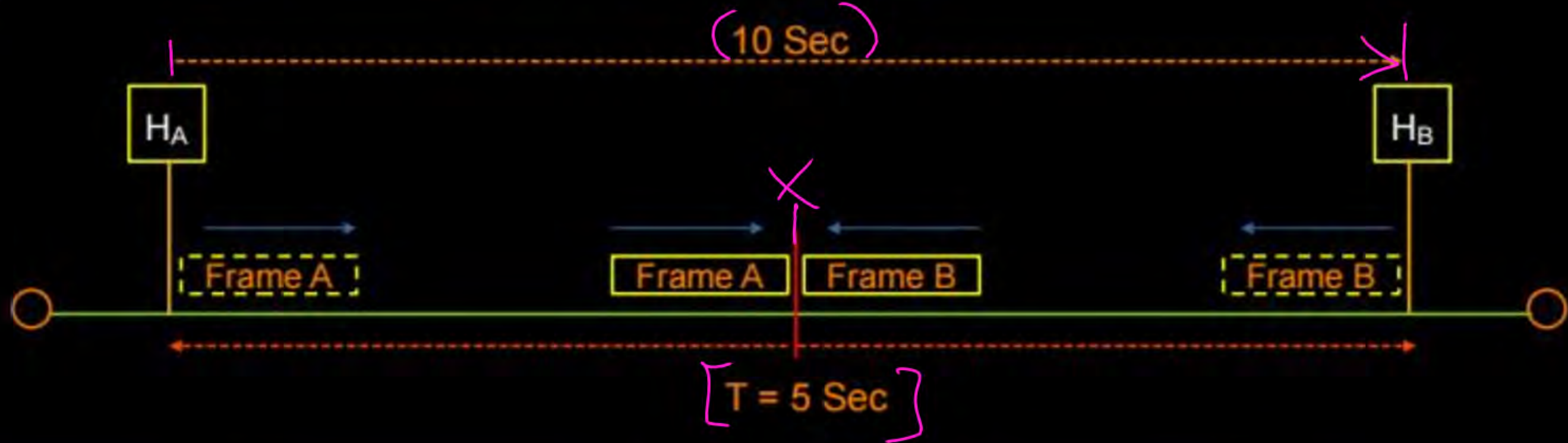
[Sense the channel, before transmission]

=> if channel sensed idle : “transmit entire frame”

=> if channel sensed busy : “defer transmission”

CASE I :-

Frame Transmission Time = 1 Sec



Host A

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0$ Sec

Complete transmission at time $T = 1$ Sec

Receive collision signals at time $T = 10$ Sec

Host B

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0$ Sec

Complete transmission at time $T = 1$ Sec

Receive collision signals at time $T = 10$ Sec

CASE II :-

Frame Transmission Time = 1 Sec

(10 Sec)



Host A

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0$ Sec

Complete transmission at time $T = 1$ Sec

Receive collision signals at time $T = 16$ Sec

Host B

Sense channel for transmission at time $T = 6$

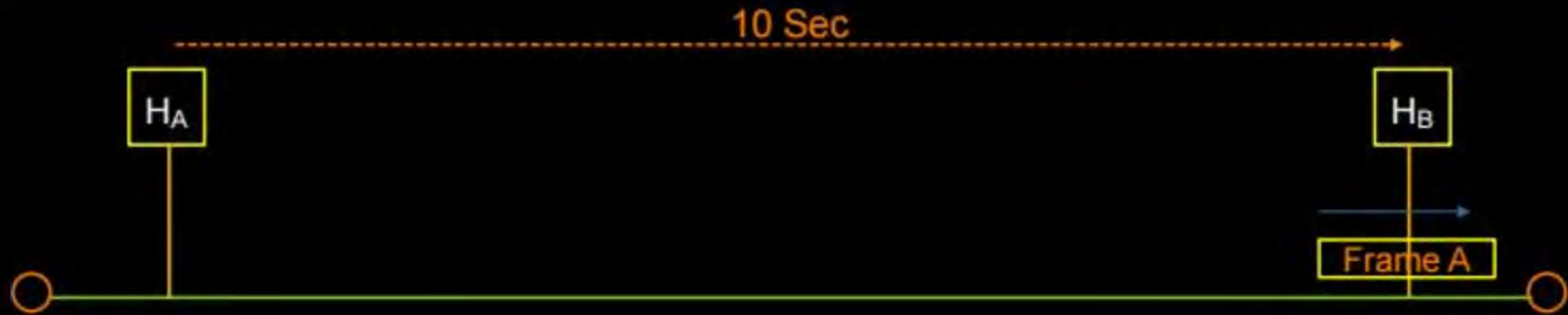
Start transmission at time $T = 6$ Sec

Complete transmission at time $T = 7$ Sec

Receive collision signals at time $T = 10$ Sec

CASE III :-

Frame Transmission Time = 1 Sec



Host A

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0$ Sec

Complete transmission at time $T = 1$ Sec

Not Receive any collision signals

Host B

Sense channel for transmission at time $T = 10.5$

Transmission defer as it find channel is busy



Topic : CSMA/CD



→ CSMA with Collision Detection

→ Sense before transmit and Sense while (during) transmission



Topic : CSMA/CD



→ Applicable only for wired LAN (Bus topology)

→ No any feedback (acknowledgment) from receiver

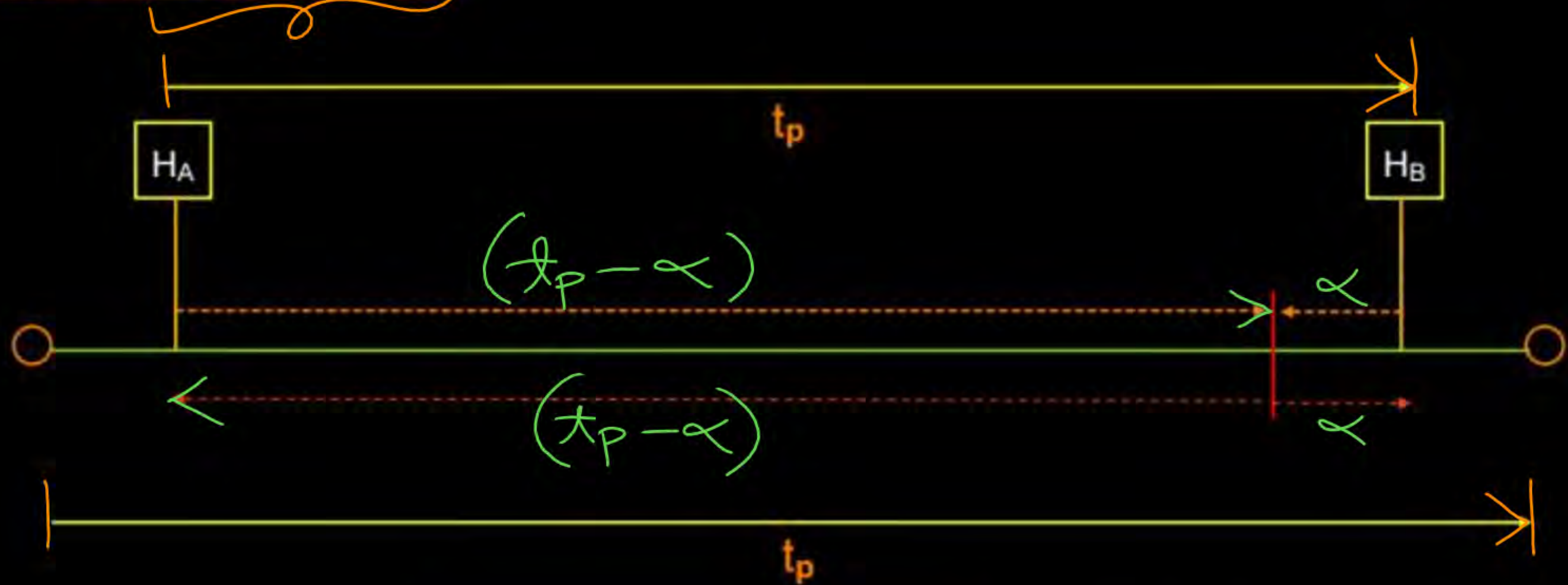
[Unlike ALOHA, CSMA & CSMA/CA]



⇒ CSMA/CD is preferable for small propagation delay network



Topic : CSMA/CD



$$t_x \geq 2(t_p - \alpha)$$

$\alpha \rightarrow$ consider negligible

$$t_x \geq 2t_p$$



Topic : CSMA/CD



→ To detect collision,
minimum frame transmission delay should be
greater than equal to (maximum) round trip propagation delay.

frame transmission delay $\geq$ round trip propagation delay

$$t_x \geq 2 * t_p$$



$$t_x = \frac{\text{Frame Size}}{\text{Bandwidth}}$$

$$t_p = \frac{\text{Distance}}{\text{Signal speed}}$$

CASE IV :-

Frame Transmission Time = 20 Sec



Host A

Sense channel for transmission at time $T = 0$

Start transmission at time $T = 0 \text{ Sec}$

It will complete transmission at $T = 20 \text{ Sec}$

Receive collision signals at time $T = 16 \text{ Sec}$

Host B

Sense channel for transmission at time $T = 6$

Start transmission at time $T = 6 \text{ Sec}$

It will complete transmission at $T = 26 \text{ Sec}$

Receive collision signals at time $T = 10 \text{ Sec}$



Topic : CSMA/CD



For minimum frame size

$$t_x = 2 * t_p$$

$$\text{minimum frame size} = (2 * t_p) * \text{Bandwidth}$$

[MCQ]

IIT-M

[GATE-2003] [2 Mark]



#Q. A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?

- ☒ A 50 bytes
- ☒ B 100 bytes
- ☒ C 200 bytes
- ☒ D None of the above

Ans: D

Solution :



Minimum frame size = $2 * (\text{Distance} / \text{Signal Speed}) * \text{Bandwidth}$

$$= 2 * \left(\frac{2 \text{ KM}}{2 * 10^8 \text{ m/sec}} \right) * 10^7 \text{ bits/sec}$$

$$= 2 * \left(\frac{2 * 10^3 \text{ m}}{2 * 10^8 \text{ m/sec}} \right) * 10^7 \text{ bits/sec}$$

$$= 200 \text{ bits} \leftarrow$$

$$= 25 \text{ bytes}$$

[MCQ]

11T-B

[GATE-IT-2005] [2 Mark]



#Q. A network with CSMA/CD protocol in the MAC layer is running at 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be

- ☐ A 5000 bits
- ☐ B 5000 bytes
- ☐ C 10000 bits
- ☐ D 10000 bytes

[MCQ]

115C

[GATE-IT-2008] [2 Mark]



#Q. The minimum frame size required for a CSMA/CD based computer network running at 1 Gbps on a 200 m cable with a link speed of 2×10^8 m/s is :

- ☐ A 125 bytes
- ☐ B 250 bytes
- ☐ C 500 bytes
- ☐ D None of these

[NAT]

115<

[GATE-2016] [2 Mark]



#Q. A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

[MCQ]

117-K

[GATE-2015] [1 Mark]



#Q. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

- A 8000
- B 10000
- C 16000
- D 20000



Topic : Ethernet

⇒ LAN standard



→ IEEE 802.3

→ Based on 1 – persistent CSMA/CD

[NAT]

115C

[GATE-2024] [2 Mark]



#Q. Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (in bits) required for collision detection is _____.

[MCQ]

11T-B

[GATE-2013] [2 Mark]



#Q. Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

- ☐ A 1
- ☐ B 2
- ☐ C 2.5
- ☐ D 5



Topic : Jam Signal



- Jamming signal 32-bits or 48-bits
- Transmitter transmit jam signal when collision detected
- To inform the other transmitting stations about the collision

$$t_x \geq [2 * t_p + t_{xj}]$$

t_{xj} : Transmission time for Jamming Signal

$$t_{xj} = \frac{\text{jam signal size}}{\text{Bandwidth}}$$

$$\text{Min}^m \text{ transmission time} = [2t_p + t_{x_j}]$$



If jam signal size or jam signal transmission time given

$$t_x \geq [2t_p + t_{x_j}]$$

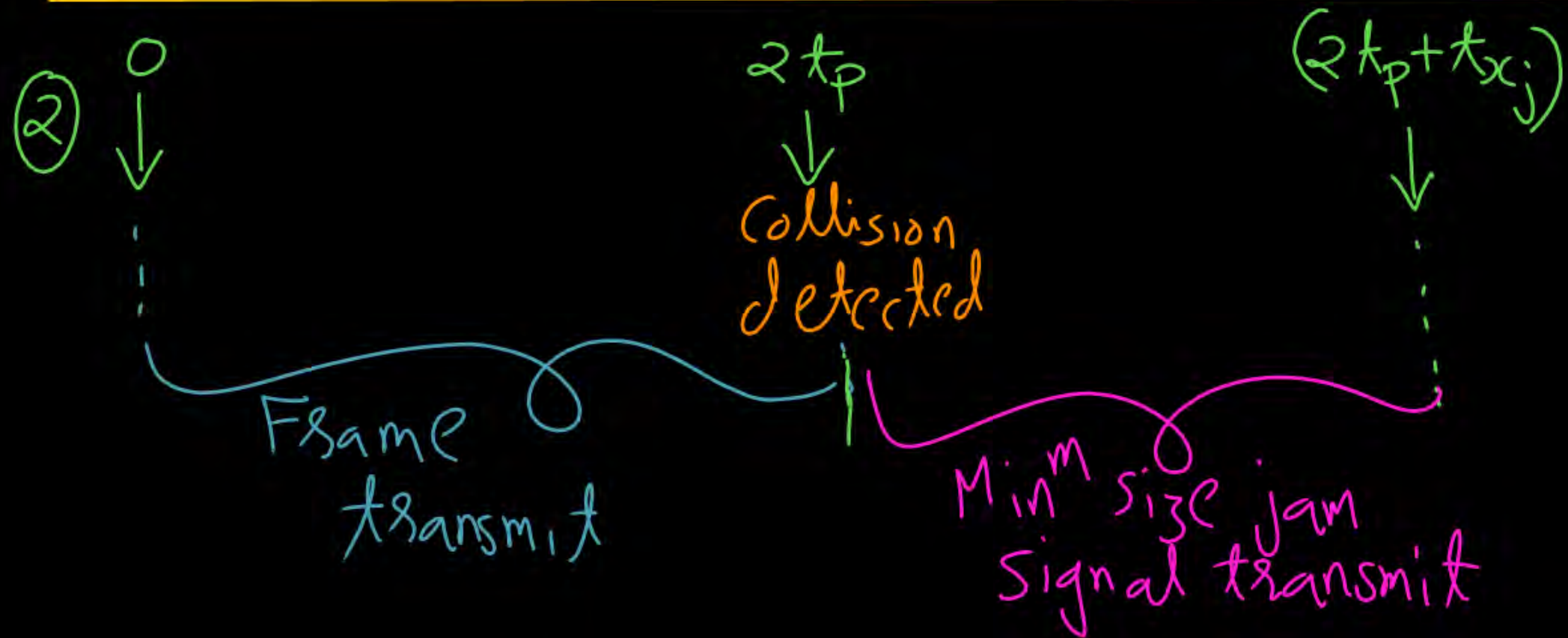
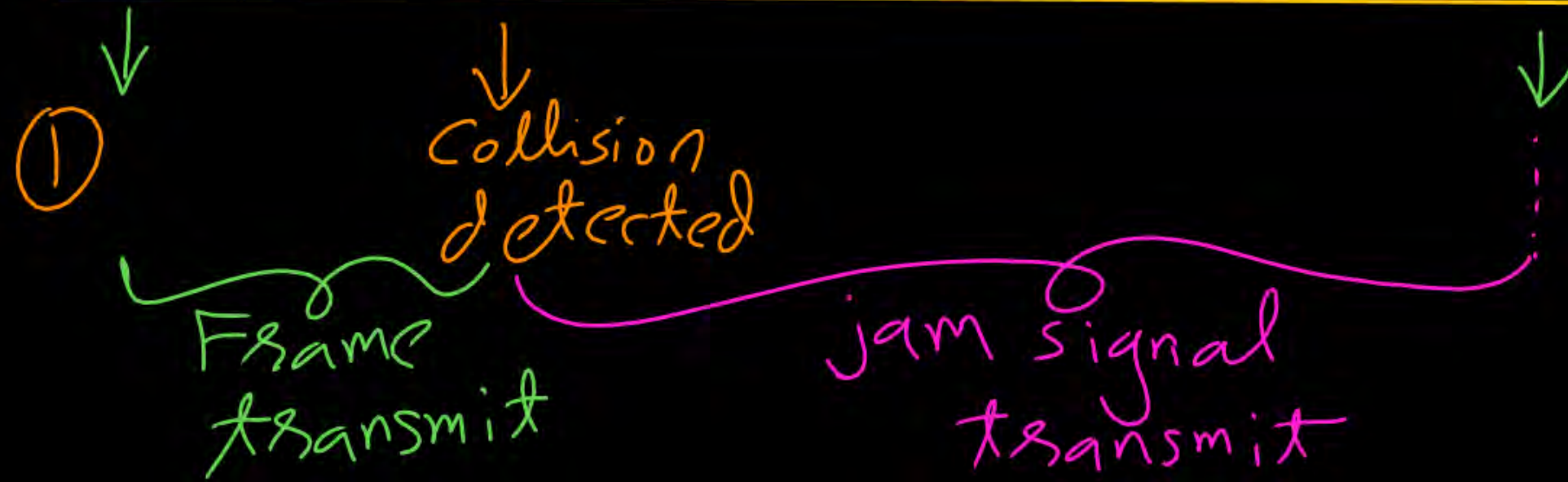
If no any info given about jam signal in the question

→ consider negligible (t_{x_j})

$$\text{Min}^m \text{ transmission time} = 2t_p$$

$$t_x \geq 2t_p$$

0 $(2t_p)$ $(2t_p + t_{xj})$ t_x



[MCQ]

IIT-B, H.W.

[GATE-CS-2005][2 Mark]



#Q. Suppose the round trip propagation delay for a 10 Mbps Ethernet having 48-bit jamming signal is 46.4 microseconds. The minimum frame size is :

- ☐ A 94 bits
- ☐ B 416 bits
- ☐ C 464 bits
- ☐ D 512 bits



2 mins Summary



Topic

CSMA ✓

Topic

CSMA/CD ✓

Topic

Ethernet →



THANK - YOU

